Sampling Examples

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Video Screenshots

Sampling Examples

If $x(t) \xleftrightarrow{FT} X(\Omega)$, $x[n] = x(nT) \xleftrightarrow{FT} X_s(\Omega)$

$$X_s(\Omega) = \frac{1}{T} \sum_{k=-\infty}^{\infty} X(\Omega - k\Omega_s) \quad ; \Omega_s = \frac{2\pi}{T}$$

Sampling Thm: If max freq in $x(t)$ is B, then

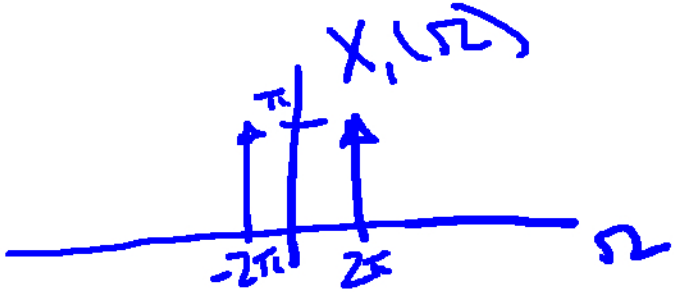
$\Omega_s > 2B$ for unique reconstruction of $x(t)$ from $x[n]$

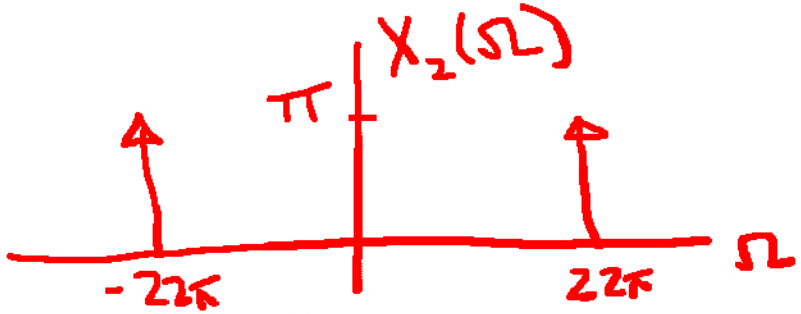
① CD quality audio max freq 20 kHz

Sample @ > 40 kHz

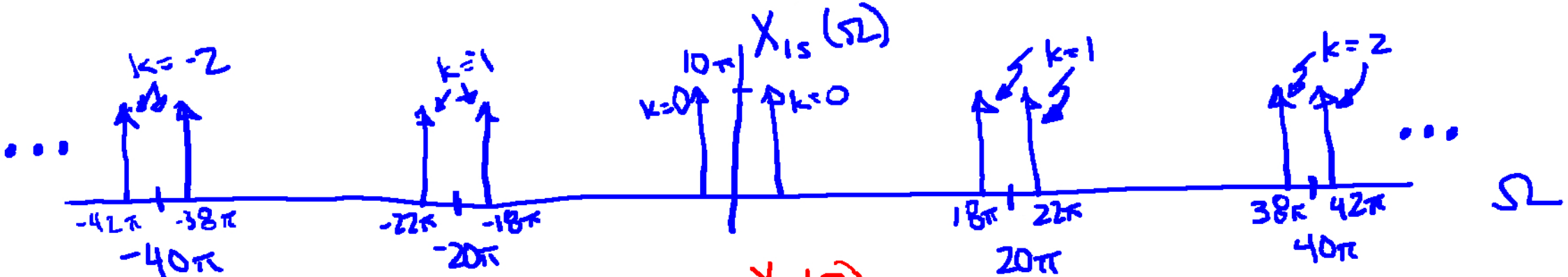
CD sampling rate : 44.1 kHz

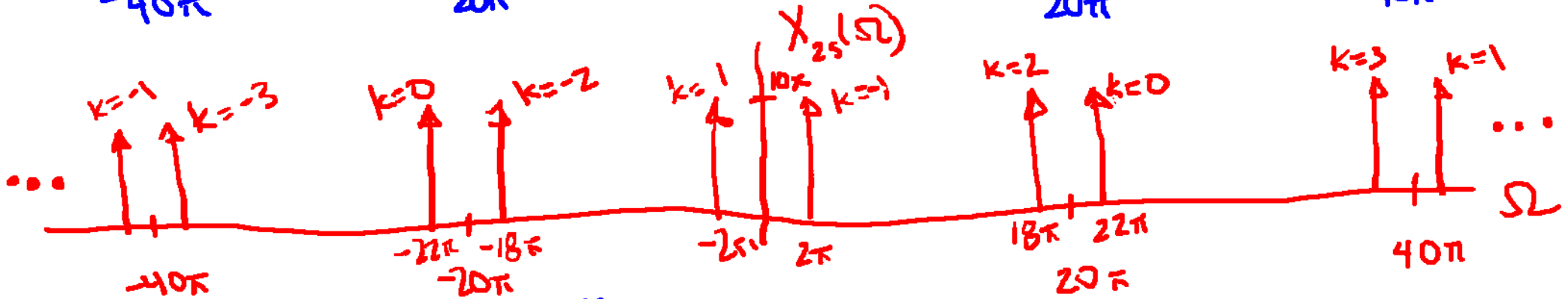
② Sinusoids $x_1(t)=\cos(2\pi t)$, $x_2(t)=\cos(22\pi t)$, $T=0.1$ sec





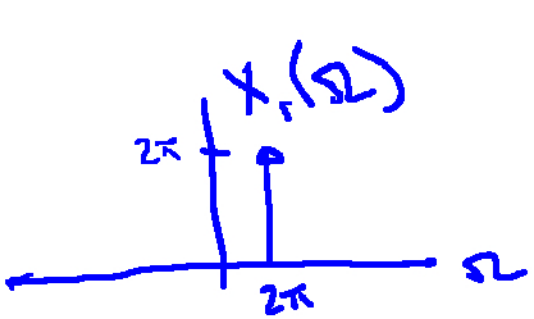
$\Omega_s = \frac{2\pi}{0.1} = 20\pi$

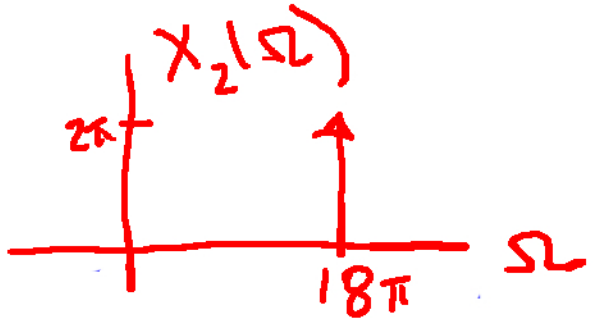




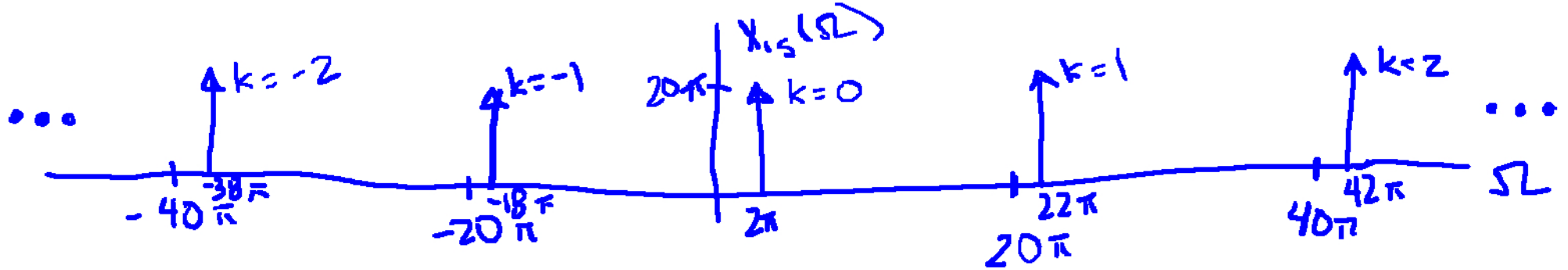
Aliasing

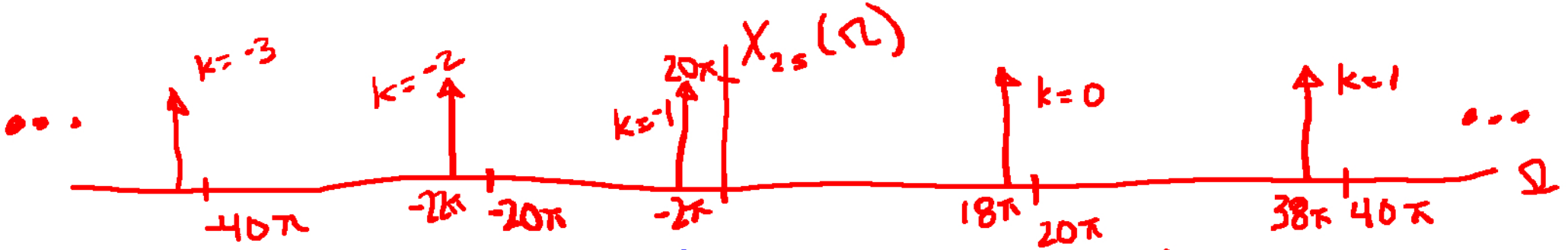
(3) Complex sinusoid $x_1(t) = e^{j2\pi t}$, $x_2(t) = e^{j18\pi t}$, $T = 0.1$ sec





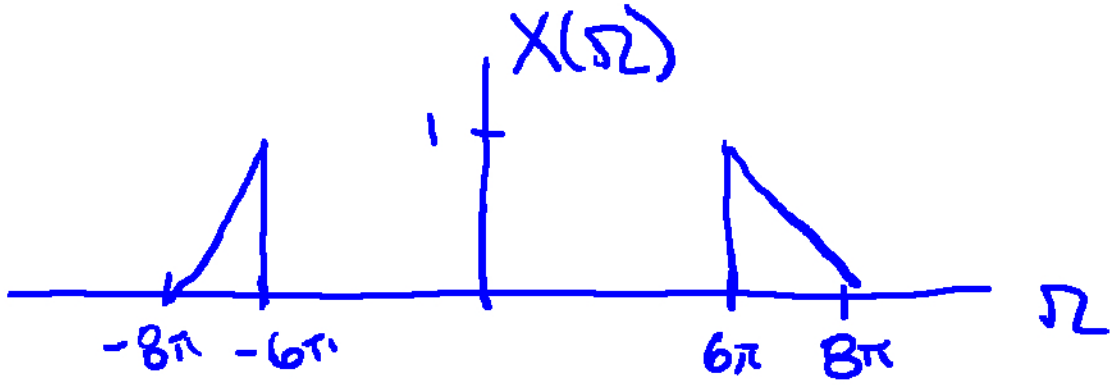
$\Omega_s = \frac{2\pi}{0.1} = 20\pi$



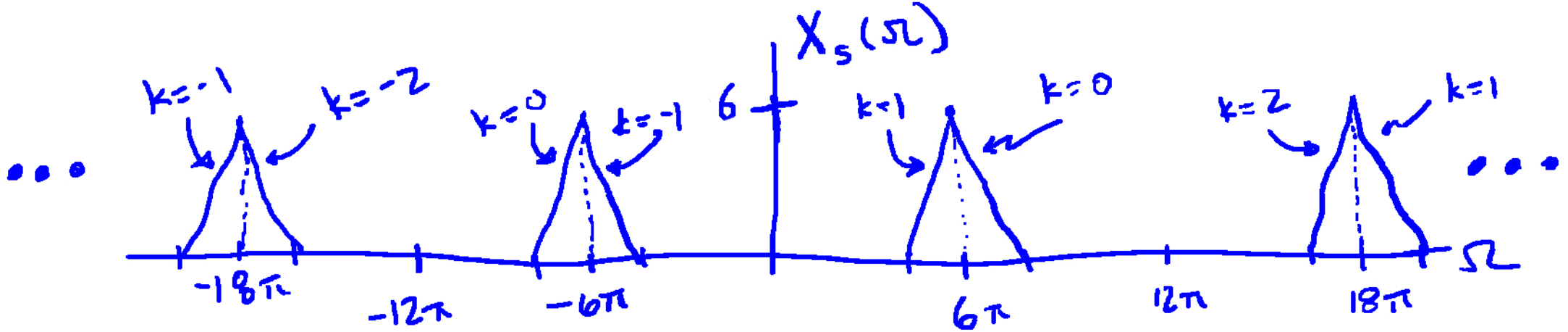


Can't distinguish $e^{j18\pi t}$ and $e^{-j2\pi t}$

④



Sample $x(t)$ at $T = 1/6\,\text{sec} \Rightarrow \Omega_s = 12\pi$



Aliasing.

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